Kubernetes Basics – Experiment Report

1. Introduction

The purpose of this experiment is to become familiar with the fundamental concepts and operations of Kubernetes by following the official tutorial 'Kubernetes Basics'. Through hands-on practice, the experiment helps learners understand how Kubernetes manages containerized applications, schedules resources, and ensures reliability and scalability.

2. Objectives

The major objectives include understanding Kubernetes components, setting up a cluster, deploying an application, exploring resources, exposing the application, scaling workloads, and updating deployments.

3. Environment Setup

The environment requires a machine capable of running Kubernetes, a local or remote cluster, kubectl tools, internet access, and screenshot tools.

4. Experiment Procedure

The experiment consists of creating a cluster, deploying an application, exploring resources, exposing the application, scaling the application, and updating the deployment.

4.1 Creating the Kubernetes Cluster

A Kubernetes cluster is initialized and verified to ensure it is ready to schedule workloads. [Insert Screenshot]

4.2 Deploying the Application

An example application is deployed and verified to be running. [Insert Screenshot]

4.3 Exploring the Application

Details of Pods, Deployments, and logs are reviewed. [Insert Screenshot]

4.4 Exposing the Application

A Service is created to expose the application externally. [Insert Screenshot]

4.5 Scaling the Application

The workload is scaled to more replicas. [Insert Screenshot]

4.6 Updating the Application

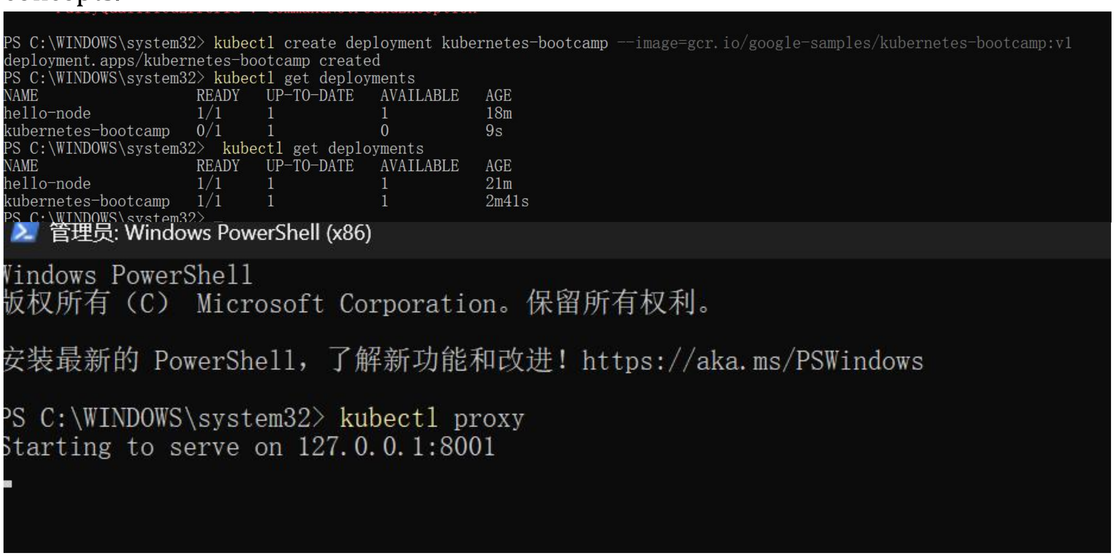
A rolling update is performed to introduce a new version. [Insert Screenshot]

5. Results and Discussion

The experiment demonstrates Kubernetes capabilities: orchestration, scaling, and rolling updates.

6. Conclusion

The experiment provides foundational Kubernetes experience and prepares learners for advanced cloud-native
concepts.

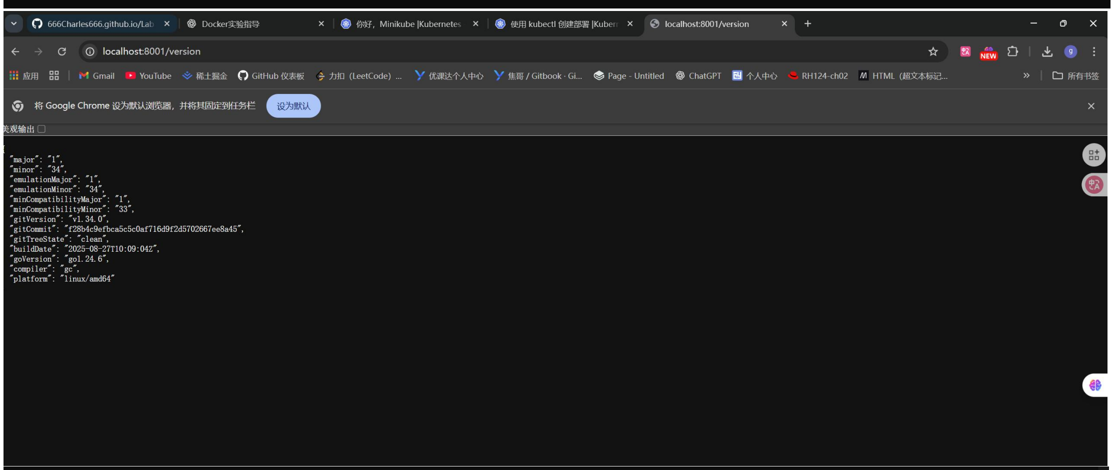


```
>> Write-Output "Name of the Pod: $POD_NAME"
>>
Name of the Pod: hello-node-6c9b5f4b59-trl74
PS C:\WINDOWS\system32> curl http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME:8080/proxy/ hello-node-6c9b5f4b59-trl74
Invoke-WebRequest : 找不到接受实际参数“hello-node-6c9b5f4b59-trl74”的位置形式参数。
所在位置 行:1 字符: 1
+ curl http://localhost:8001/api/v1/namespaces/default/pods/$POD_NAME:8 ...
+
    + CategoryInfo          : InvalidArgument: (:) [Invoke-WebRequest], ParameterBindingException
    + FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.InvokeWebRequestCommand

PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> curl http://localhost:8001/api/v1/namespaces/default/pods/hello-node-6c9b5f4b59-trl74:8080/proxy/


StatusCode        : 200
StatusDescription : OK
Content           : NOW: 2025-11-18 09:47:27.059125025 +0000 UTC m=+1660.230169283
RawContent        : HTTP/1.1 200 OK
                    Audit-Id: 673c2aa2-a517-46cf-b492-e10edb009aa2
                    Content-Length: 62
                    Cache-Control: no-cache, private
                    Content-Type: text/plain; charset=utf-8
                    Date: Tue, 18 Nov 2025 09:47:27 GMT

                    N...
Forms             : {}
Headers           : {[Audit-Id, 673c2aa2-a517-46cf-b492-e10edb009aa2], [Content-Length, 62], [Cache-Control, no-cache,
                    private], [Content-Type, text/plain; charset=utf-8]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 62
```




```
PS C:\WINDOWS\system32> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
hello-node-6c9b5f4b59-trl74           1/1     Running   0          28m
kubernetes-bootcamp-658f6cbd58-xb7vc  1/1     Running   0          10m
PS C:\WINDOWS\system32> kubectl describe pods
Name:             hello-node-6c9b5f4b59-trl74
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 18 Nov 2025 17:19:24 +0800
Labels:           app=hello-node
                  pod-template-hash=6c9b5f4b59
Annotations:      <none>
Status:           Running
IP:               10.244.0.6
IPs:
  IP:           10.244.0.6
Controlled By:  ReplicaSet/hello-node-6c9b5f4b59
Containers:
  agnhost:
    Container ID:  docker://84706cd1055a365757238c153b8944da2dd945caedcba5ce3f13c1bc76198bcc
    Image:         registry.k8s.io/e2e-test-images/agnhost:2.53
    Image ID:      docker-pullable://registry.k8s.io/e2e-test-images/agnhost@sha256:99c6b4bb4a1e1df3f0b3752168c89358794d02258ebebc26bf21c29399011a85
    Port:          <none>
    Host Port:     <none>
    Command:
      /agnhost
      netexec
      --http-port=8080
    State:          Running
      Started:      Tue, 18 Nov 2025 17:19:46 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-nz4cx (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-nz4cx:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  29m   default-scheduler  Successfully assigned default/hello-node-6c9b5f4b59-trl74 to minikube
  Normal  Pulling    29m   kubelet            Pulling image "registry.k8s.io/e2e-test-images/agnhost:2.53"
```

```
Name:             kubernetes-bootcamp-658f6cbd58-xb7vc
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 18 Nov 2025 17:37:49 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=658f6cbd58
Annotations:      <none>
Status:           Running
IP:               10.244.0.8
IPs:
  IP:           10.244.0.8
Controlled By:  ReplicaSet/kubernetes-bootcamp-658f6cbd58
Containers:
  kubernetes-bootcamp:
    Container ID:   docker://4ba08681a5b07d2ab30d56f1d1babe131a4c4d254e0c04cf1e872a1c4df23e35
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
    Image ID:       docker-pullable://gcr.io/google-samples/kubernetes-bootcamp@sha256:0d6b8ee63bb57c5f5b6156f446b3bc3b3c143d233037f3a2f00e279c8fcc64af
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Tue, 18 Nov 2025 17:39:32 +0800
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-bckm4 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-bckm4:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  10m   default-scheduler  Successfully assigned default/kubernetes-bootcamp-658f6cbd58-xb7vc to minikube
  Normal  Pulling    10m   kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:v1"
  Normal  Pulled     9m3s  kubelet            Successfully pulled image "gcr.io/google-samples/kubernetes-bootcamp:v1" in 1m42.078s (1m42.078s including waiting). Image size: 211336459 bytes.
  Normal  Created    9m2s  kubelet            Created container: kubernetes-bootcamp
  Normal  Started    9m2s  kubelet            Started container kubernetes-bootcamp
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl exec "hello-node-6c9b5f4b59-trl74" -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=hello-node-6c9b5f4b59-trl74
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
HOME=/root
PS C:\WINDOWS\system32> kubectl exec -ti hello-node-6c9b5f4b59-trl74 -- bash
hello-node-6c9b5f4b59-trl74:/# cat server.js
cat: can't open 'server.js': No such file or directory
hello-node-6c9b5f4b59-trl74:/# ls -al
total 96512
drwxr-xr-x    1 root     root          4096 Nov 18 09:19 .
drwxr-xr-x    1 root     root          4096 Nov 18 09:19 ..
-rwxr-xr-x    1 root     root             0 Nov 18 09:19 .dockerenv
```

```
hello-node-6c9b5f4b59-trl74:/# curl http://localhost:8080
NOW: 2025-11-18 09:59:49.384022469 +0000 UTC m=+2402.614727905hello-node-6c9b5f4b59-trl74:/#
```

```
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
hello-node-6c9b5f4b59-trl74            1/1     Running   0          42m
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running   0          23m
PS C:\WINDOWS\system32> kubectl get services
NAME         TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
hello-node   LoadBalancer   10.104.239.130   <pending>     8080:32475/TCP   34m
kubernetes   ClusterIP      10.96.0.1        <none>        443/TCP          48m
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
service/kubernetes-bootcamp exposed
PS C:\WINDOWS\system32> kubectl describe services/kubernetes-bootcamp
Name:                     kubernetes-bootcamp
Namespace:                default
Labels:                   app=kubernetes-bootcamp
Annotations:              <none>
Selector:                 app=kubernetes-bootcamp
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.246.186
IPs:                      10.100.246.186
Port:                     <unset>  8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset>  31398/TCP
Endpoints:                10.244.0.8:8080
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
PS C:\WINDOWS\system32> 
```

```
PS C:\WINDOWS\system32> minikube service kubernetes-bootcamp --url
http://127.0.0.1:59867
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```

管理员: Windows PowerShell (x86)

```
    Command:
      /agnhost
      netexec
      --http-port=8080
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
  Node-Selectors: <none>
  Tolerations:    <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   hello-node-6c9b5f4b59 (1/1 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  52m   deployment-controller  Scaled up replica set hello-node-6c9b5f4b59 from 0 to 1


Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Tue, 18 Nov 2025 17:37:49 +0800
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=kubernetes-bootcamp
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:         gcr.io/google-samples/kubernetes-bootcamp:v1
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-658f6cbd58 (1/1 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  34m   deployment-controller  Scaled up replica set kubernetes-bootcamp-658f6cbd58 from 0 to 1
PS C:\WINDOWS\system32> kubectl get pods -l app=kubernetes-bootcamp
NAME                                   READY   STATUS    RESTARTS   AGE
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running   0          34m
```

```
666Charles666.github.io/Lab    使用服务曝光您的应用 |Kuber    查看文件不存在原因    127.0.0.1:65448
127.0.0.1:65448
应用  Gmail  YouTube  稀土掘金  GitHub 仪表板  力扣 (LeetCode) ...  优课达个人中心  焦哥 / Gitbook · Gi...  Page - Untitled  ChatGPT  个人中心  RH124-ch02
将 Google Chrome 设为默认浏览器，并将其固定到任务栏  设为默认

Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-658f6cbd58-xb7vc | v=1
```

```
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath="{.items[0].metadata.name}")
PS C:\WINDOWS\system32> echo "Name of the Pod: $POD_NAME"
Name of the Pod: hello-node-6c9b5f4b59-trl74
PS C:\WINDOWS\system32>
```

```
hello-node-6c9b5f4b59-trl74
PS C:\WINDOWS\system32> echo "Name of the Pod: $POD_NAME"
Name of the Pod:
PS C:\WINDOWS\system32> $POD_NAME = (kubectl get pods -o jsonpath="{.items[0].metadata.name}")
PS C:\WINDOWS\system32> echo "Name of the Pod: $POD_NAME"
Name of the Pod: hello-node-6c9b5f4b59-trl74
PS C:\WINDOWS\system32> kubectl label pods "$POD_NAME" version=v1
pod/hello-node-6c9b5f4b59-trl74 labeled
PS C:\WINDOWS\system32> kubectl describe pods "$POD_NAME"
Name:             hello-node-6c9b5f4b59-trl74
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 18 Nov 2025 17:19:24 +0800
Labels:           app=hello-node
                  pod-template-hash=6c9b5f4b59
                  version=v1
Annotations:      <none>
Status:           Running
IP:               10.244.0.6
IPs:
  IP:           10.244.0.6
Controlled By:  ReplicaSet/hello-node-6c9b5f4b59
Containers:
  agnhost:
    Container ID:  docker://84706cd1055a365757238c153b8944da2dd945caedcba5ce3f13c1bc76198bcc
    Image:         registry.k8s.io/e2e-test-images/agnhost:2.53
    Image ID:      docker-pullable://registry.k8s.io/e2e-test-images/agnhost@sha256:99c6b4bb4a1e1df3f0b3752168c89358794d02258ebebc26bf21c29399011a85
    Port:          <none>
    Host Port:     <none>
    Command:
      /agnhost
      netexec
      --http-port=8080
    State:          Running
      Started:      Tue, 18 Nov 2025 17:19:46 +0800
```

```
  Normal  Started    54m   kubelet              Started container agnhos
PS C:\WINDOWS\system32> kubectl get pods -l version=v1
NAME                          READY   STATUS    RESTARTS   AGE
hello-node-6c9b5f4b59-trl74   1/1     Running   0          55m
PS C:\WINDOWS\system32>
```

```
NAME                          READY   STATUS    RESTARTS   AGE
hello-node-6c9b5f4b59-trl74   1/1     Running   0          55m
PS C:\WINDOWS\system32> kubectl delete service -l app=kubernetes-bootcamp
service "kubernetes-bootcamp" deleted
PS C:\WINDOWS\system32> kubectl get services
NAME         TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
hello-node   LoadBalancer   10.104.239.130   <pending>     8080:32475/TCP   48m
kubernetes   ClusterIP      10.96.0.1        <none>        443/TCP          61m
PS C:\WINDOWS\system32> curl http://"$(minikube ip):$NODE_PORT"
```

```
PS C:\WINDOWS\system32> kubectl exec -ti $POD_NAME -- curl http://localhost:8080
NOW: 2025-11-18 10:16:39.065281852 +0000 UTC m=+3412.383714046
```

```
PS C:\WINDOWS\system32> kubectl exec -ti $POD_NAME -- curl http://localhost:8080
NOW: 2025-11-18 10:16:39.065281852 +0000 UTC m=+3412.383714046
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="LoadBalancer" --port 8080
service/kubernetes-bootcamp exposed
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
hello-node            1/1     1            1           58m
kubernetes-bootcamp   1/1     1            1           39m
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl get rs
NAME                             DESIRED   CURRENT   READY   AGE
hello-node-6c9b5f4b59            1         1         1       58m
kubernetes-bootcamp-658f6cbd58   1         1         1       39m
PS C:\WINDOWS\system32> kubectl get rs
NAME                             DESIRED   CURRENT   READY   AGE
hello-node-6c9b5f4b59            1         1         1       58m
kubernetes-bootcamp-658f6cbd58   1         1         1       39m
PS C:\WINDOWS\system32> kubectl scale deployments/kubernetes-bootcamp --replicas=4
deployment.apps/kubernetes-bootcamp scaled
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
hello-node            1/1     1            1           58m
kubernetes-bootcamp   4/4     4            4           40m
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl get pods -o wide
NAME                                   READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
hello-node-6c9b5f4b59-tr174            1/1     Running   0          58m   10.244.0.6    minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-5cv8p   1/1     Running   0          14s   10.244.0.9    minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-5ws9v   1/1     Running   0          14s   10.244.0.11   minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-7wztl   1/1     Running   0          14s   10.244.0.10   minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running   0          40m   10.244.0.8    minikube   <none>           <none>
PS C:\WINDOWS\system32> kubectl describe deployments/kubernetes-bootcamp
Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Tue, 18 Nov 2025 17:37:49 +0800
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=kubernetes-bootcamp
Replicas:               4 desired | 4 updated | 4 total | 4 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:         gcr.io/google-samples/kubernetes-bootcamp:v1
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
```

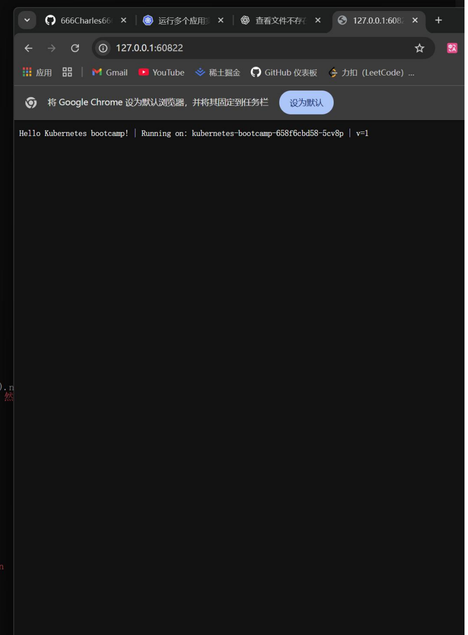
```
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> kubectl get pods -o wide
NAME                                   READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
hello-node-6c9b5f4b59-tr174            1/1     Running   0          58m   10.244.0.6    minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-5cv8p   1/1     Running   0          14s   10.244.0.9    minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-5ws9v   1/1     Running   0          14s   10.244.0.11   minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-7wztl   1/1     Running   0          14s   10.244.0.10   minikube   <none>           <none>
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running   0          40m   10.244.0.8    minikube   <none>           <none>
PS C:\WINDOWS\system32> kubectl describe deployments/kubernetes-bootcamp
Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Tue, 18 Nov 2025 17:37:49 +0800
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=kubernetes-bootcamp
Replicas:               4 desired | 4 updated | 4 total | 4 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:         gcr.io/google-samples/kubernetes-bootcamp:v1
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-658f6cbd58 (4/4 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  40m   deployment-controller  Scaled up replica set kubernetes-bootcamp-658f6cbd58 from 0 to 1
  Normal  ScalingReplicaSet  22s   deployment-controller  Scaled up replica set kubernetes-bootcamp-658f6cbd58 from 1 to 4
PS C:\WINDOWS\system32>
```

管理员: Windows PowerShell (x86)

```
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-658f6cbd58 (4/4 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  40m   deployment-controller  Scaled up replica set kubernetes-bootcamp-658f6cbd58 from 0 to 1
  Normal  ScalingReplicaSet  22s   deployment-controller  Scaled up replica set kubernetes-bootcamp-658f6cbd58 from 1 to 4
PS C:\WINDOWS\system32> kubectl describe services/kubernetes-bootcamp
>>
Name:                     kubernetes-bootcamp
Namespace:                default
Labels:                   app=kubernetes-bootcamp
Annotations:              <none>
Selector:                 app=kubernetes-bootcamp
Type:                     LoadBalancer
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.111.43.189
IPs:                      10.111.43.189
Port:                     <unset>  8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset>  31594/TCP
Endpoints:                10.244.0.8:8080,10.244.0.10:8080,10.244.0.9:8080 + 1 more...
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
PS C:\WINDOWS\system32> export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-template='{{(index .spec.ports 0).n
export : 无法将“export”项识别为 cmdlet、函数、脚本文件或可运行程序的名称。请检查名称的拼写，如果包括路径，请确保路径正确，然
所在位置 行:1 字符: 1
+ export NODE_PORT="$(kubectl get services/kubernetes-bootcamp -o go-te ...
+ ~~~~~~
    + CategoryInfo          : ObjectNotFound: (export:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\system32> echo NODE_PORT=$NODE_PORT
NODE_PORT=
PS C:\WINDOWS\system32> $NODE_PORT = (kubectl get service kubernetes-bootcamp -o jsonpath="{.spec.ports[0].nodePort}")
PS C:\WINDOWS\system32> echo "NODE_PORT=$NODE_PORT"
NODE_PORT=31594
PS C:\WINDOWS\system32> curl http://"$(minikube ip):$NODE_PORT"
curl : 无法连接到远程服务器
所在位置 行:1 字符: 1
+ curl http://"$(minikube ip):$NODE_PORT"
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-WebRequest], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeWebRequestCommand

PS C:\WINDOWS\system32> minikube service kubernetes-bootcamp --url
http://127.0.0.1:60822
! 因为你正在使用 windows 上的 Docker 驱动程序，所以需要打开终端才能运行它。
```




```
PS C:\WINDOWS\system32> ^C
PS C:\WINDOWS\system32> kubectl scale deployments/kubernetes-bootcamp --replicas=2
deployment.apps/kubernetes-bootcamp scaled
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
hello-node            1/1     1            1           62m
kubernetes-bootcamp   2/2     2            2           43m
PS C:\WINDOWS\system32> kubectl get pods -o wide
>>
NAME                                   READY   STATUS    RESTARTS   AGE     IP            NODE       NOMINATED NODE   RE
ADINESS GATES
hello-node-6c9b5f4b59-trl74            1/1     Running   0          62m     10.244.0.6    minikube   <none>           <n
one>
kubernetes-bootcamp-658f6cbd58-5ws9v   1/1     Running   0          4m16s   10.244.0.11   minikube   <none>           <n
one>
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running   0          44m     10.244.0.8    minikube   <none>           <n
one>
PS C:\WINDOWS\system32> _
```

管理员: Windows PowerShell (x86)

```
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-bckm4 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-bckm4:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  45m   default-scheduler  Successfully assigned default/kubernetes-bootcamp-658f6cbd58-xb7vc to minikube
  Normal  Pulling    45m   kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:v1"
  Normal  Pulled     43m   kubelet            Successfully pulled image "gcr.io/google-samples/kubernetes-bootcamp:v1" in 1m42.078s (1m42.078s including waiting). Image size: 211336459 bytes.
  Normal  Created    43m   kubelet            Created container: kubernetes-bootcamp
  Normal  Started    43m   kubelet            Started container kubernetes-bootcamp
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kubernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS              RESTARTS   AGE
hello-node-6c9b5f4b59-trl74            1/1     Running             0          63m
kubernetes-bootcamp-57cc954bb9-wx6jh   0/1     ContainerCreating   0          13s
kubernetes-bootcamp-658f6cbd58-5ws9v   1/1     Running             0          5m29s
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running             0          45m
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kubernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS              RESTARTS   AGE
hello-node-6c9b5f4b59-trl74            1/1     Running             0          63m
kubernetes-bootcamp-57cc954bb9-wx6jh   0/1     ContainerCreating   0          13s
kubernetes-bootcamp-658f6cbd58-5ws9v   1/1     Running             0          5m29s
kubernetes-bootcamp-658f6cbd58-xb7vc   1/1     Running             0          45m
PS C:\WINDOWS\system32> kubectl expose deployment/kubernetes-bootcamp --type="NodePort" --port 8080
Error from server (AlreadyExists): services "kubernetes-bootcamp" already exists
PS C:\WINDOWS\system32> $NODE_PORT = kubectl get service kubernetes-bootcamp -o jsonpath="{.spec.ports[0].nodePort}"
PS C:\WINDOWS\system32> echo "NODE_PORT=$NODE_PORT"
NODE_PORT=31594
PS C:\WINDOWS\system32> _
```

```
PS C:\WINDOWS\system32> kubectl rollout status deployments/kubernetes-bootcamp
>>
deployment "kubernetes-bootcamp" successfully rolled out
PS C:\WINDOWS\system32> kubectl describe pods
```

```
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=gcr.io/google-samples/kubernetes-bootcamp:v10
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32> kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
hello-node            1/1     1            1           68m
kubernetes-bootcamp   2/2     1            2           50m
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS         RESTARTS   AGE
hello-node-6c9b5f4b59-trl74            1/1     Running        0          68m
kubernetes-bootcamp-57cc954bb9-8b2r4   1/1     Running        0          4m12s
kubernetes-bootcamp-57cc954bb9-wx6jh   1/1     Running        0          5m2s
kubernetes-bootcamp-677ff875c4-t24g9   0/1     ErrImagePull   0          12s
```

管理员: Windows PowerShell (x86)

```
Start Time:       Tue, 18 Nov 2025 18:27:52 +0800
Labels:           app=kubernetes-bootcamp
                  pod-template-hash=677ff875c4
Annotations:      <none>
Status:           Pending
IP:               10.244.0.14
IPs:
  IP:           10.244.0.14
Controlled By:  ReplicaSet/kubernetes-bootcamp-677ff875c4
Containers:
  kubernetes-bootcamp:
    Container ID:
    Image:          gcr.io/google-samples/kubernetes-bootcamp:v10
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ImagePullBackOff
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-chw6k (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-chw6k:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason     Age                From               Message
  ----     ------     ----               ----               -------
  Normal   Scheduled  29s                default-scheduler  Successfully assigned default/kubernetes-bootcamp-677ff875c4-t24g9 to minikube
  Normal   BackOff    25s                kubelet            Back-off pulling image "gcr.io/google-samples/kubernetes-bootcamp:v10"
  Warning  Failed     25s                kubelet            Error: ImagePullBackOff
  Normal   Pulling    14s (x2 over 29s)  kubelet            Pulling image "gcr.io/google-samples/kubernetes-bootcamp:v10"
  Warning  Failed     11s (x2 over 26s)  kubelet            Failed to pull image "gcr.io/google-samples/kubernetes-bootcamp:v10": Error response from daemon: manifest for gcr.io/google-samples/kubernetes-bootcam
p:v10 not found: manifest unknown: Failed to fetch "v10"
  Warning  Failed     11s (x2 over 26s)  kubelet            Error: ErrImagePull
PS C:\WINDOWS\system32> kubectl rollout undo deployments/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp rolled back
PS C:\WINDOWS\system32> kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
hello-node-6c9b5f4b59-trl74            1/1     Running   0          69m
kubernetes-bootcamp-57cc954bb9-8b2r4   1/1     Running   0          4m43s
kubernetes-bootcamp-57cc954bb9-wx6jh   1/1     Running   0          5m33s
PS C:\WINDOWS\system32> kubectl delete deployments/kubernetes-bootcamp services/kubernetes-bootcamp
deployment.apps "kubernetes-bootcamp" deleted
service "kubernetes-bootcamp" deleted
PS C:\WINDOWS\system32> _
```